

Task 01 – Problem Definition and Literature Review

1. Problem Definition

This project aims to create and evaluate a machine learning model to categorize patients according to three risk levels (low, medium, high), as represented by the target variable `disease_risk_level`. The model will perform a multi-class supervised classification task. The model will analyze patients based on different inputs, including (but not limited to) demographic data (e.g. age, sex, and blood type), clinical records (e.g. BP, blood sugar, cholesterol, and BMI), hospital administration (e.g. department of the patient's admission, disease history of patient, department of treatment requested/lab tests used) and economic factors (e.g. consultation, accommodation, lab tests, and medicines involved).

Identifying disease risk early is a priority in preventive healthcare: it allows doctors to intervene before a patient gets into an acute situation and lets a hospital assign its scarce resources consultation hours, laboratory capabilities, hospital beds to patients in the most need. In particular, SmartCare's automated risk classifier helps with three concrete goals: enabling prioritization of high-risk patients and more timely appointments; giving doctors a data-based second opinion to go along with clinical judgment instead of replacing it; and helping the hospital understand how intensity and costs of treatment relate to patient risk profiles.

In contrast to a binary risk-prediction task, the presence of a three-class ordinal target not only adds complexities to modelling practices but also likely affects the contents of this report including class imbalance among the three risk types; use of multi-class evaluation metrics instead of binary metrics; and as will be demonstrated in the subsequent sections of this report, differences in model behaviour owing to linear separability of the risk categories in the feature space.

1.2 Significance

- Supports preventive care rather than purely reactive treatment.
- Helps triage clinical staff time and hospital resources.
- Assists capacity planning(e.g. anticipating room/ICU needs for high-risk cohorts).
- Provides a decision-support layer, not a replacement for clinical judgement.

2. Literature Review

Overall, this review uses a total of seven peer-reviewed and published sources two more sources than the minimum number of five sources needed for this coursework throughout its seven sections of studies about heart disease prediction, infectious disease risk, risk prediction benchmark in healthcare, multi-task risk classification, ordinal classification of countries based on the level of risk, and multi-class severity of disease. Collectively, the cited sources include basic binary prediction series that are mostly cited in healthcare AI studies and the smaller literature that treats the risk of the disease just like option C and calls it a multi-class problem.

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2.1 Existing AI Approaches and Common Algorithms

The predominant method in the reviewed literature is supervised, tabular machine learning on structured clinical or demographic data. Mavrogiorgou et al. compare seven classical algorithms (Naïve Bayes, KNN, Decision Tree, Logistic Regression, Random Forest, Neural Networks, and Stochastic Gradient Descent) for six healthcare risk scenarios, concluding that no algorithm prevails for all risk scenarios. In cases where the target is clearly multi-class, Trigka, Dritsas and Mylonas compare One-vs-All and One-vs-One decomposition approaches for four-level severity of anaemia, with Logistic Regression used under the One-vs-One strategy performing best (~95% accuracy). Suma et al. (2024) train natively multi-class classifiers (Random Forest, SVM, and Logistic Regression) for a three-level COPD risk score, with Random Forest having the best performance (92% accuracy). In a relevant methodological aspect, Bird et al. convert continuous risk at the level of country into four ordinal classes using the K% binning method, training stacked ensembles. Banerjee and Paçal and Liu, Liu and Liu review a large amount of research, reaffirming the predominance of classical ML and ensemble approaches in structured healthcare risk predictions, reserving deep learning for the case of image or signal inputs.

2.2 Current Limitations and Research Gaps

There are four recurring limitations throughout the studies discussed in this article which will influence the approach taken in this research: (i) firstly, the existing research investigating disease risk often provides results in terms of binary variables, rather than multi-class variables, and examples of research utilizing multi-class classification with a dataset as complex as that of SmartCare are rare (ii) Secondly, many studies with the research designs in multi-class disease severity fail to deal with class imbalance, where “too many subjects are wrongly classified as severe anaemia” (Trigka, Dritsas and Mylonas, 2023). (iii) Thirdly, lack of generalizability is very typical of this type of research, which is affirmed by Banerjee and Paçal (2025). (iv) Lastly, model transparency still remains among the weaknesses of effective ensemble models, in which high-performing models are used but, still, understanding of their working principles is poor, which is why Explainable AI was introduced in this research.